

Technology Stack

Date	30 June 2026
Team ID	SWTID-2026-3278
Project Name	Credit Card Approval Prediction
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS, JavaScript	Provides a user-friendly interface for entering applicant details and displaying the prediction result.

2	Backend / Server-Side	Python (Flask)	Processes user requests, runs the machine learning model, and returns the approval prediction.
3	Database / Data Storage	CSV/ Pandas	Stores applicant information, prediction history, and user data for future reference.
4	Cloud / Hosting / Deployment	Scikit-learn, Joblib	<i>Used to train the Credit Card Approval Prediction model and save it for deployment.</i>
5	Version Control & CI/CD	Git, GitHub	Tracks project changes, supports collaboration, and maintains version history.

6	Third-Party APIs / Other Tools	Pandas, NumPy, Matplotlib	Pandas handles data preprocessing, NumPy performs numerical computations, and Matplotlib visualizes data.
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